DYS-I-CAN: An Aid for the Dyslexic to improve the skills using Mobile Application

Sarah I [1], Soundarya K [2],Tamil Thendral S[3], Dhanalakshmi R [4]and Deenadayalan T[5]
Department of Computer Science and Engineering
sarahirfan3199@gmail.com, soundaryakishore05@gmail.com,
thentral98@gmail.com, dhanalakshmisai@gmail.com, deenadayalan.cse@kcgcollege.com

***Abstract*— Dyslexia – A Learning disability is a medical learning disorder which involves difficulties in reading, word decryption, comprehension, memory, writing, spelling, and speaking. People diagnosed with learning disability exhibits lack of self-esteem, eager and anxiety as they are not able to mingle with the society. This paper describes a mobile application that helps the dyslexic users affected with reading, writing difficulties in real world. This example will show the text such as alphabets and numbers around them and read it loudly. Additionally the utilization of a machine learning approach to boost the effectiveness of the education in a dyslexic child is needed.**

Keywords—Mobile Application, Dyslexia, Machine learning

I. INTRODUCTION

Dyslexia may be a specific issue in learning, reading, orthography and writing. Since reading may be a key component in learning, youngsters with learning disability will have hassle mastering basic skills and succeeding at school. They have issues processing the knowledge when they see a word. Typically a dyslexic kid can have hassle connecting the sound created by a particular letter or deciphering the sounds of all the letters along with a word of that kind. Given these challenges, youngsters with learning disability even have hassle with writing, spelling, and arithmetic.The appropriate learning scheme ought to be designed to produce learners up to date. This learning pattern consists of technology and pedagogy. Pedagogy is the approach of teaching and learning that has to suit the learners, and what additional to those learning, disabled youngsters.

Apps are one in all the technologies which will be accustomed to ease reading difficulties, science calculations and communication issues. This mobile application provides audio for alphabets and numbers. This step will assist them to differentiate the dissimilarities between letters and numbers. Random alphabets are going to be generated for assessment of the dyslexic student. Spellings aretaught and are evaluated too.The objective is to assist dyslexic youngsters to acknowledge and distinguish letter and numbers like p, q, b, d, m, and w… and a pair of, 3, 5, 6, 7, 9… to assist them learn higher.

II. BACKGROUND

Few topics are going to be enclosed in this section: related work, learning mobile apps, Pedagogy for Recognizing and identifying Letters, numbers.

A. Applications of dyslexia

- Youngsters with learning disability are having acquisition disorders and this makes it troublesome in mastering the ability to scan, spell and write. The objective for the primary version of 'Dyslexia Baca' is to help alphabet recognition; thereupon they're driven to find out and might recall the knowledge in an exceedingly fascinating method**.** The letters and numbers of alphabets areshown within the balloons. The primary version introduced 5 confusing letters (P, p, b, d and m, n) for dyslexic learners at early reading level[4]

- This analysis study analyses impact of E-Learning Tool named R-U-LEXIC for dyslexic youngsters with special respect to selective state, in the South Asian country. The approach involves activity wherever youngsters could also be screened on a mass scale for learning disability. It proposes Assessments to Teachers, students and children based on Age or Academic level. The Assessment allocates marks which may help to know the understanding level of the dyslexic student.

- Several applications offered during this domain facilitate them by correcting their problems by enjoying games and reading books. This correcting method takes time and dyslexics become helpless once addressing their daily activities. The method will help to read the text loudly so user will be allowed to customise the text according to

 ISBN 978-1-7281-6202-7

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 13:03:58 UTC from IEEE Xplore. Restrictions apply.

their. The system provides the user with a chance to beat the reading difficulties by merely using camera possibility within the developed application. Then the mobile application can establish the texts so that it can be read loudly (Text-To-Speech)[1]

B. Mobile Applications – Technology Enabling Platform

The ease of use of Technology with the advent of Mobile Applications (Apps) is developed using packages and are accessible using devices like smart phones and tablets. Due to the new advances in technology Pedagogy in Teaching Learning Environment have become more simple and easy applicable. It also helps the dyslexic students to learn more efficiently and quickly based on their necessary skills and learning capability. It also helps them to assess their learning capability and improve further. Accordingly it is also very important to know the students interest and goals to make them understand and improve. It is also very essential to have kindle their interest by rewarding them with incentives and acquire parental involvement

C. Identifying and Distinguishing Letters / numbers

There exist various challenges in improving the teaching Pedagogy of the dyslexic students as they encounter the following difficulties

- Alphabet recognition
- Remembering the alphabet name
- Adaption over technology based learning like mobile apps.
- Difficulty in reading
- Difficulty in understanding maths
- Difficulty in oral instruction involving multiple directions

Technology enabled applications like Smartphone based learning using mobile applications, voice recognition, sound replication and various other tools that are helpful and will increase the performance in dyslexic youngsters.

III. "DYS-I CAN"- AN AID FOR THE DYSLEXIC USING MOBILE APPLICATION

DYS I-CAN is the mobile application developed for dyslexic students. Few objectives that can be achieved over dyslexic students can be listed as follows

i) Aid to recognize alphabet

ii) Enabling the dyslexic children learn to read;

iii) to improve the learning capability in recalling the data that they read and learn.

After completing every level, dyslexic children are motivated through incentives and rewards which will kindle their interest to move to further levels

Table 1 : LEVELS IN DYS I-CAN APP

LEVEL 1	LEVEL 2	LEVEL 3
ENGLISH	MATHS	COMMUNICATION [text to speech]
SPELLING	NUMBERS	LISTENING
COLOUR IDENTIFICATION	SIMPLE OPERATIONS	SPEAKING

This mobile app consists of modules and every module has several levels. Dyslexic students will use this game to create learning fun crammed and amusing. Thevarious modules are English, math, listening, speaking and writing. Every module focuses specifically on each component,in English alphabets, then random alphabets generation then word buildings and color identification. Of these things can effectively increase the capacity on the person. Further modules are explained below in detail.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 13:03:58 UTC from IEEE Xplore. Restrictions apply.

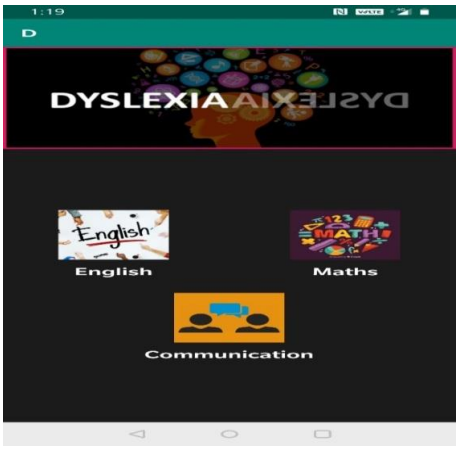


Fig 1. Modules in DYS- I-CAN APP

A. *English module*

This module consists of two sublevels that are orthography and color identification. Before stepping into the orthography sublevel the alphabets are instructed with audio and till they're clear with the alphabets they'll use it multiple times to find out clearly and also to assess themselves. The color identification sublevel teaches the colors and assesses them

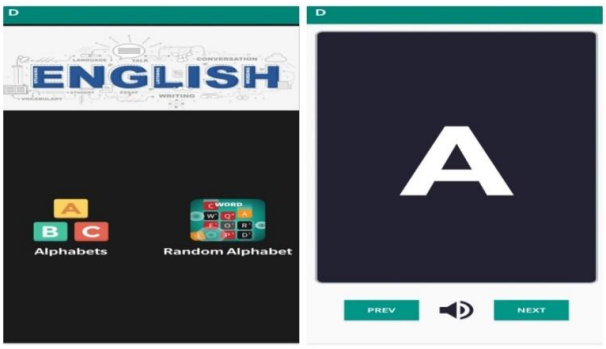


Fig 2. DYS-I-CAN English Module

B. *Maths Module*

This module consists of two sublevels that are numbers and operations. Before stepping into the operations sublevel, the numbers are instructed with audio and till they are clear with the numbers they will use it multiple times to assess themselves.

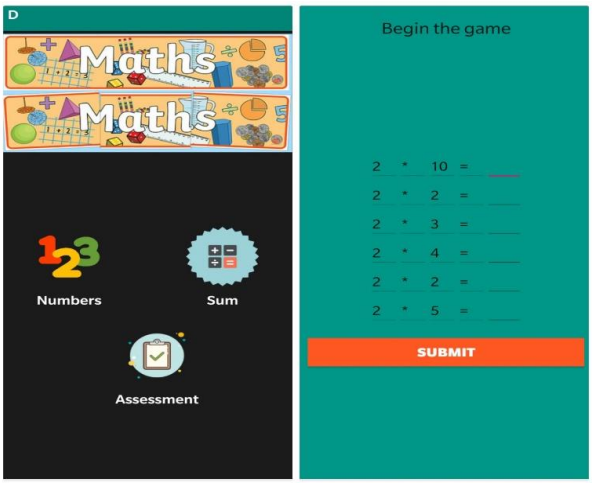


Fig 3. DYS-I-CAN Maths Module

C . *Communication Module*

This module consists of 2 sublevels which are listening and speaking. This module can be attained or evaluated only after understanding the English Module

IV. ALGORITHM

A. *Neural Machine Translation algorithm*

Natural Language Processing involves various tasks like text summarization, speech recognition, etc. and they can be attained using Sequence-to-Sequence (seq2seq) models. The aim is to translate given sentences from English to Tamil. Neural Machine Translation (NMT) is similar to the encoder decoder structure which accepts any input and converts it to the respected output. NMT model is to take a sentence in one language as input and return that sentence translated into a distinct language as output. For e.g Language Translation from English to Hindi or English to Spanish

B. *Steps involved*

NMT is the encoder decoder structure. A sequence to sequence model lies behind numerous systems which is appreciated in everyday lives. For instance, seq2seq model powers applications like Google Translate, voice-enabled devices and online chatbots.

There are steps involved as follows

- Get input in the variety of text

ISBN 978-1-7281-6202-7
Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 13:03:58 UTC from IEEE Xplore. Restrictions apply.

- Create a vocabulary to assign distinctive index to each distinctive word as similar as to vocabulary for every language would merely contain every unique word in that language
- Each word is transformed into a vector where every word within the dataset is represented numerically
- Every vector is to assign an index to every distinctive word within the input language and hence do the above steps repetitively for the output language

a	0
the	1
red	2
orange	3
blue	4
black	5
fish	6
whale	7
beaver	8
ate	9
drank	10
<SOS>	11
<EOS>	12

$$the = \begin{bmatrix}0\\1\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\end{bmatrix} \quad blue = \begin{bmatrix}0\\0\\0\\0\\1\\0\\0\\0\\0\\0\\0\\0\end{bmatrix} \quad whale = \begin{bmatrix}0\\0\\0\\0\\0\\0\\0\\1\\0\\0\\0\\0\end{bmatrix} \quad ate = \begin{bmatrix}0\\0\\0\\0\\0\\0\\0\\0\\1\\0\\0\\0\end{bmatrix} \quad the = \begin{bmatrix}0\\1\\0\\0\\0\\0\\0\\0\\0\\0\\0\\0\end{bmatrix} \quad red = \begin{bmatrix}0\\0\\1\\0\\0\\0\\0\\0\\0\\0\\0\\0\end{bmatrix} \quad fish = \begin{bmatrix}0\\0\\0\\0\\0\\0\\1\\0\\0\\0\\0\\0\end{bmatrix}$$

In order to understand the Encoder Decoder rule which is to be implemented in seq2seq model the Encoder portion of the model takes a sentence within the input language and creates a concept vector from this sentence. This thought vector stores the meaning of the sentence which is passed to a Decoder that outputs the interpretation of the sentence within the output language.

The working o NMT is well depicted in the figures as follows

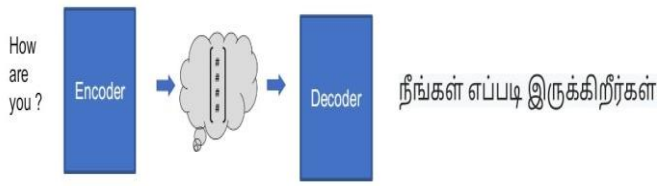


Fig 4. Neural Machine Translation

At every time intervals every word within the input sentence is fed severally into the model of Neural Machine Translation during defined time intervals or every time step . During the execution and at every time-step, t, the model updates a hidden vector, h, information from the word inputted to the model at that time-step. This hidden vector works to store information regarding the inputted sentence.

At certain time period when no words have been inputted to the Encoder at time-step t=0, the hidden state within the Encoder has the tendency to starts out as Associate in Nursing empty which have an inclination to represent this hidden state. During the time t = 1, the hidden state in the Encoder points to corresponding state in Decoder which is the corresponding output state. Hence at time step t =1 , the encoder can point to the respective state of the output and updates in hidden state. In order to match the input words to the output vocabulary the Decoder can use an additional weight matrix or threshold to create likelihood over all of the words.

B. Architecture

Dyslexic students may have problems in reading and in comprehending the given text. He/She may also experience difficulty in understanding Maths and hence in order to improve the learning skills the DYS –I-CAN Mobile App is designed. It comprises of various models with English, Communication and Maths. English Module helps to recognize alphabets, Spell check and identifying colours. Maths Module helps to recognize numbers and performing simple arithmetic operations like Addition and subtraction

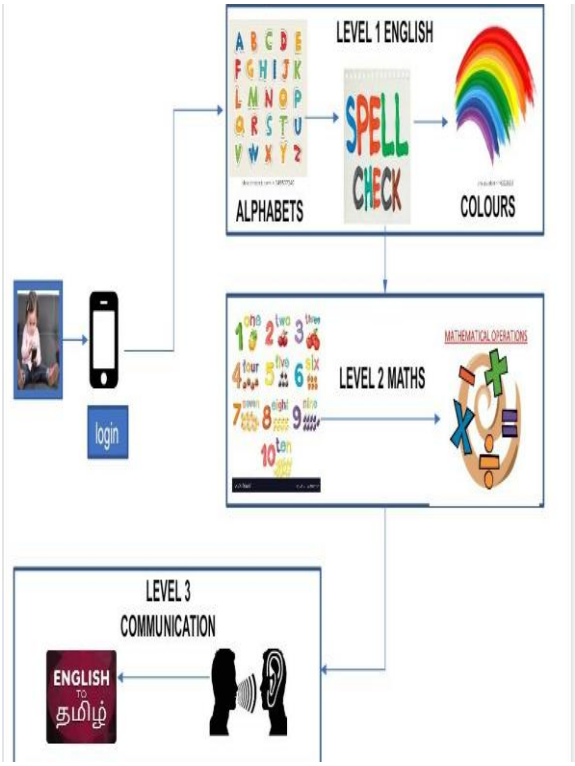


Fig 5. Architecture Diagram – DYS-I-CAN

ISBN 978-1-7281-6202-7
Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 13:03:58 UTC from IEEE Xplore. Restrictions apply.

B. Results and performance

The performance of the dyslexic student can be evaluated using the scores that will be generated at the end of the levels. This app can be used for dyslexic children between the age 3 to 12

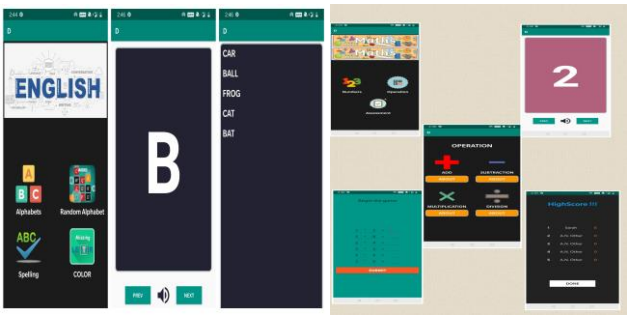


Fig 5. Learning Using English and Maths Module

V. CONCLUSION

It is highly essential to know the difficulties of the dyslexic students and to attempt the learning disorders through advancement of Technology. The major challenge of dyslexic students is character Identification or recognition which is very important for various activities such as spell, read and write. It is very important for any individual to recognize, memorize, learn repeatedly.

The existing method adopted in classrooms is monotonous, uninteresting and boring to the learners. But with the emerging advancements in technology it is essential to improve the learning interest of any individual especially to the needy like dyslexic student. The guidance and support of teacher and parents through the mobile application helps to improve the existing methodology and will ease the process of education.

Mobile Apps to know and improve the requirements for dyslexic children helps to visualize alphabets on screen and to listen to those letters in the form of audio. The main objective is to help the dyslexic students in alphabet recognition which will help them to remember and recall the learnt data in and interesting and exciting way. In addition to that it helps to learn numbers and perform simple arithmetic operations. Further to the above simple models , it also helps the dyslexic children to improve their communication skills

References

[1] Aziz, Fakhrul & Husni, Husniza & Jamaludin, Zulikha. (2013). Translating Interaction Design Guidelines for Dyslexic Children's Reading Application. Lecture Notes in Engineering and Computer Science. 2. 977-980.

[2] Kalhoro, Quratulain & Dhirani, Lubna & Abbasi, Tamoor & Abbasi, Shazia. (2010). M-learning -an innovative advancement of ICT in education. 148 - 151. 10.1109/ICDLE.2010.5606017.

[3] Kast, Monika & Meyer, Martin & Voegeli, Christian & Gross, Markus & Jäncke, Lutz. (2007). Computer-based multisensory learning in children with developmental dyslexia. Restorative neurology and neuroscience. 25. 355-69.

[4] Mohd Daud, Salwani & Abas, Hafiza. (2013). 'Dyslexia Baca' Mobile App -- The Learning Ecosystem for Dyslexic Children. Proceedings - 2013 International Conference on Advanced Computer Science Applications and Technologies, ACSAT 2013. 412-416. 10.1109/ACSAT.2013.87.

[5] N. B. Mohamad *et al*., "Spectral analysis based brain imaging of normal, poor dyslexic, and capable dyslexic children in reading, writing and spelling task," *2014 IEEE Conference on Biomedical Engineering and Sciences (IECBES)*, Kuala Lumpur, 2014, pp. 653-656.

[6] S. Mohamad, W. Mansor and K. Y. Lee, "Review of neurological techniques of diagnosing dyslexia in children," *2013 IEEE 3rd International Conference on System Engineering and Technology*, Shah Alam, 2013, pp. 389-393. doi: 10.1109/ICSEngT.2013.6650206

[7] Shaywitz, Sally & Shaywitz, Bennett. (2005). Dyslexia (Specific Reading Disability). Biological psychiatry. 57. 1301-9. 10.1016/j.biopsych.2005.01.043.

[8] S. Rajapakse, D. Polwattage, U. Guruge, I. Jayathilaka, T. Edirisinghe and S. Thelijjagoda, "ALEXZA: A Mobile Application For Dyslexics Utilizing Artificial Intelligence And Machine Learning Concepts," *2018 3rd International Conference on Information Technology Research (ICITR)*, Moratuwa, Sri Lanka, 2018, pp. 1-6.

[9] V. Thulasibai, P. M. B. Devamalar, B. SenthilKumar and A. Marx, "An empirical study of the impact of E-Learning Tool developed for dyslexic children with special reference to selective schools in Tamil Nadu, South India," *2016 2nd International Conference on Advances in Electrical, Electronics, Information, Communication and Bio-Informatics (AEEICB)*, Chennai, 2016, pp. 752-760.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 11,2025 at 13:03:58 UTC from IEEE Xplore. Restrictions apply.